

Global Facilities - Design & Construction Standard – Electrical - On-Site Solar Panel Installation

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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

- 1.1 This document serves as a System Standard document that provides specifications and guidelines that shall be followed during the Design, Engineering and Construction of new Solar Photovoltaic (PV) system
- 1.2 This document is a Micron Specification which provides the general principles to be followed in the electrical system and equipment engineering design, construction and testing/commissioning. It applies to all Greenfield projects. This document is a principle reference for brown field revamp and rejuvenation projects.
- 1.3 Design and engineering of the electrical installation shall satisfy all statutory requirements of the national and/or local authorities of the country in which the installation is to be located.
- 1.4 The electrical installation shall be suitable for the site conditions as specify by the owner.
- 1.5 The contents of this specification and of standards and publications referred herein shall be adhered to, except where amended by specific requirements provide by owner of related installation and is allowed under statutory requirements mentioned above.
- 1.6 Any contradiction between the requirements of this specifications and IEC or national standards shall be subject to clarification and approval by the Owner.

- 1.7 Standardize equipment of similar nature or identical materials/components and of similar or identical construction should be from the same Manufacturer.
- 1.8 The Contractor shall obtain from the Manufacturer's test reports in accordance with the equipment specifications.

2.0 Scope

- 2.1 The document covers the new Base build System installations only.
- 2.2 Site (s) Impacted:
This standard specifies requirements and provides recommendations for the design and construction of On-Site Solar Panel in Micron. It applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.
- 2.3 For Power Purchase Agreement (PPA) related projects, any deviations from this document shall be subject to clarification and approval by the Owner and Micron

3.0 Definitions and Acronyms

Term/Acronym	Definition/Description
The Contractor	The party that carries out all or part of the engineering design, purchasing, installation, testing and commissioning of a project. The Owner may undertake all or part of the duties of the Contractor.
The Manufacturer/ Supplier	The party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.
The Owner	The party that initiates the project and eventually pays for it. The Owner may also include an agent or consultant authorized to act for, and on behalf of, the Owner.
Alternating Current (AC)	Alternating Current (AC)
Altitude	The height of the Sun from the horizon.
Azimuth	The position of the Sun in relation to the cardinal directions. The solar industry standard is to express azimuth clockwise from true north (0°–360°); however it can also be quoted with a direction, east or west (i.e. 0°–180°E or 0°–180°W).

Term/Acronym	Definition/Description
Battery	A container consisting of one or more cells, in which chemical energy is converted into electricity and used as a power source.
Cell efficiency	The fraction of electrical power produced by a solar PV cell per amount of light energy hitting the cell.
Circuit	A circuit is the path that current flows from one charged point to another.
Circuit breaker	A mechanical device which will open a circuit under fault conditions. When too much current passes through, the device will open and prevent current flow. The circuit breaker can then be manually operated to close the circuit.
Combiner box	An electrical component for combining and housing the wiring from the solar PV array.
Current	Current is the movement of charge and is measured in amperes (A). Conventional current is the 'flow' from positive to negative, and is opposite to electron flow.
Direct Current (DC)	Electricity in which the current always moves in the same direction.
DC power optimisers (DC conditioning units)	DC to DC converters connected to individual or small groups of solar PV modules, usually incorporating a maximum power point tracker.
Energy	The amount of electric energy transferred, a product of power and time. Energy is measured in watt hours (Wh)
Energy Yield	A photovoltaic system's energy output measured over a given period of time is its yield. This value is a common measurement as it is used to indicate performance as well as forming the basis of income
Equipotential bonding	Equipotential bonding (or protective earthing) is the connecting of conductive metalwork electrically to the Earth so that it is at the same voltage (potential) as Earth throughout. This is required for safety reasons to protect people from electric shocks.
Extra Low Voltage (ELV)	Electrical systems that operate under 120 V DC (ripple free) or 50 V AC. ELV systems do not require an electrical license to be installed or worked on.
Functional earthing	The electrical connection of PV positive or negative conductors to the Earth. This is designed to ensure optimal performance of the PV array, but it is required only if specified by the manufacturer.

Term/Acronym	Definition/Description
Fuse	A device that protects conductors from excessive current. The fuse is rated to carry a certain current, and when this current is exceeded the fuse will open the circuit (by melting).
Irradiation (or Insolation)	The total amount of solar radiation available per unit area over a specified time period such as one day. It is the sum of irradiance values over a time period and is often measured in kWh/m ² /year or MJ/m ² /day. It is a measure of energy.
Junction box	A box containing a junction of electric wires or cables.
Kilowatt peak (kWp)	A non-SI unit used in the solar industry to describe the nominal power of a solar PV system: it refers to the peak output of the system under standard test conditions.
Maximum Power Point (MPP)	The Solar PV module operation point where the current and voltage generated by the Solar PV module gives the maximum power. It occurs when the load resistance is equal to the internal resistance of the Solar PV cell.
Maximum Power Point Tracker (MPPT)	An electronic device included within the inverter that alters the PV array's electrical output so that it is performing at the maximum power possible at any given time.
Micro Inverter	Small inverters connected to only one or two Solar PV modules.
Module efficiency	The fraction of electrical power produced by the Solar PV module per amount of light energy hitting the module. This is typically lower than cell efficiency due to losses from reflection from glass etc.
Monocrystalline solar cells	The most efficient and most expensive solar cells available. They have a smooth monochromatic appearance.
Multimode Inverter	An inverter that is able to operate as a grid-interactive inverter as well as an off-grid inverter, supplying specific loads during disconnection from the grid.
Nominal	A nominal value is a reference value used to describe batteries, modules or systems. Therefore, it is not an exact value. For example, the nominal voltage of a 72-cell solar module is 24 V but the open circuit voltage (Voc) for the same module can be 45.6 V and the maximum power voltage (Vmp) can be 35 V.
Open Circuit	An open circuit is where the current path is broken so that the current is equal to 0.

Term/Acronym	Definition/Description
Passive Protection	The over and under voltages, and the over and under frequency trip settings in the inverter.
Photovoltaic (PV)	The process which allows a device to create electricity when sunlight hits its surface.
Power	Power is the rate at which electric energy is transferred. Power is measured in watts (W), and is calculated using $\text{Power} = \text{Voltage (V)} \times \text{Current (I)}$.
Power Conversion Equipment (PCE)	A device that converts electrical power from one kind of power to another. In the context of Solar PV systems this usually refers to an inverter.
PV array	Strings of Solar PV modules are electrically connected in parallel to form an array. Also called a solar array.
PV module	A unit consisting of Solar PV cells that are physically and electrically connected. These cells are held together by a frame and covered by a protective substance such as glass. Also called a solar module.
PV string	When Solar PV modules are connected in series they form a string.
PV system	The Solar PV array and all associated equipment required to make it work. Also called a solar electric system.
Resistance	The opposition to current; measured in ohms (Ω).
Short Circuit	A short circuit is where the current is flowing in a closed path across the source terminals.
Solar Modules	See PV module.
Standard Test Conditions (STC)	Standardized test conditions which make it possible to conduct uniform comparisons of Solar PV modules by different manufacturers.
Surge protection	Devices or appliances that are used to protect electrical devices from voltage spikes.
Switch-Disconnect (or Isolator)	Mechanical switching device capable of making, carrying and breaking currents in normal circuit conditions and, when specified, in given operating overload conditions.
Voltage	The electrical potential difference between two points. Voltage is measured in volts (V).
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs

Term/Acronym	Definition/Description
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Design & Construction Requirement

5.1 Compliance with Standards and Guidelines

The Solar PV system and related installation must be designed and installed in accordance with the most recent Standards and guidelines including, but not limited to the following :

- a. IEC 62548:2016- Photovoltaic (PV) arrays - Design Requirements
- b. IEC 61730-1:2016- Photovoltaic (PV) module safety qualification - Part 1: Requirements for construction
- c. IEC 61730-2:2016 - Photovoltaic (PV) module safety qualification - Part 2: Requirements for testing
- d. Factory Mutual (FM) Global
 - i. FMDS0115 - Roof-Mounted Solar Photovoltaics Panel
 - ii. FMDS7106 - Ground-Mounted Solar Photovoltaics Panel
- e. Underwriter's Laboratory (UL)
 - i. UL 1703: PV modules
 - ii. UL 1741: Converters, charge controllers and combiner boxes
 - iii. UL 2703: Racking systems
 - iv. UL 4703: Photovoltaic (PV) wire
 - v. UL 2579: Photovoltaic system
- f. NFPA 70 National Electric Code (NEC).
- g. NFPA 101 Life Safety Code.
- h. NFPA 318 – Standard for the Protection of Semiconductor Fabrication Facilities
- i. NFPA 70E – Standard for Electrical Safety in the Workplace
- j. NFPA 1, Fire Code
- k. NFPA 70B, Recommended Practice for Electrical Equipment Maintenance (Chapter 33)
- l. NFPA 70E, Standard for Electrical Safety in the Workplace
- m. NFPA 5000, Building Construction and Safety Code
- n. International Solar Energy Provisions (ISEP)
- o. International Building Code (IBC)
- p. Occupational Safety and Health Act Standards (OSHA).
- q. Local Code and Regulations
- r. Global Facilities - Electrical –Electrical Specification Design Standard

5.2 System Planning

5.2.1 System Provider

Unless identified otherwise by Micron, the system provider must carry out the design, supply, installation and commissioning of the Solar PV system and all necessary ancillary work to deliver the completion of the Solar PV system as specified.

Where products containing hazardous materials are proposed, e.g. Cadmium Telluride, Lithium Ion batteries etc. Micron may require that the system provide recycle or dispose of key system components at end of life. For Micron to assess this requirement, the system provider will provide recycling and/or disposal information for the Solar PV modules, Inverters and batteries (if applicable).

In carrying out the installation of solar power systems, the system provider must be required to undertake the work in accordance with the related local regulation and guideline. In addition, personnel carrying out work relating to the PV system must hold appropriate licenses, accreditations and certificates. Specifically:

- The designer of any Solar PV system must be fully accredited;
- Person undertaking installation work must have qualified installer accreditation;
- All Low Voltage electrical work must be carried out by licensed electricians or certified competent person.

5.2.2 Site Assessment

A site assessment of the installation site should be carried out prior to the installation. As a minimum, the following should be assessed:

- Suitability of building roof for system installation by taking into consideration:
 - a) Solar access e.g. roof orientation and pitch, shading from objects on roof;
 - b) Available space for Solar PV modules;
 - c) Roof condition including likelihood of repair in the next 10 years;
 - d) Structural integrity of roof structure to bear extra weight and wind loading of the proposed solar installation
 - e) Suitability of the roof as a whole for the proposed installation work, such that no damage is likely to

occur to the infrastructure during installation and maintenance of the system.

- Suitability of existing AC mains supply:
 - a) Rating of electrical distribution board;
 - b) Identify if upgrades to electrical distribution board are needed to accommodate switches and/or reduce losses;
 - c) Confirm that potential voltage rise will not exceed required values.
- Solar resource:
 - a) Ensure that there are accurate insolation data for the site: taking into account location specific factors including shading from surrounding topography and buildings and local weather patterns.
- Availability of space for installation of proposed Solar PV system components in accordance with installation requirements e.g. restriction of access to inverters and space needed for the Solar PV array.

5.2.3 Safe Roof Access

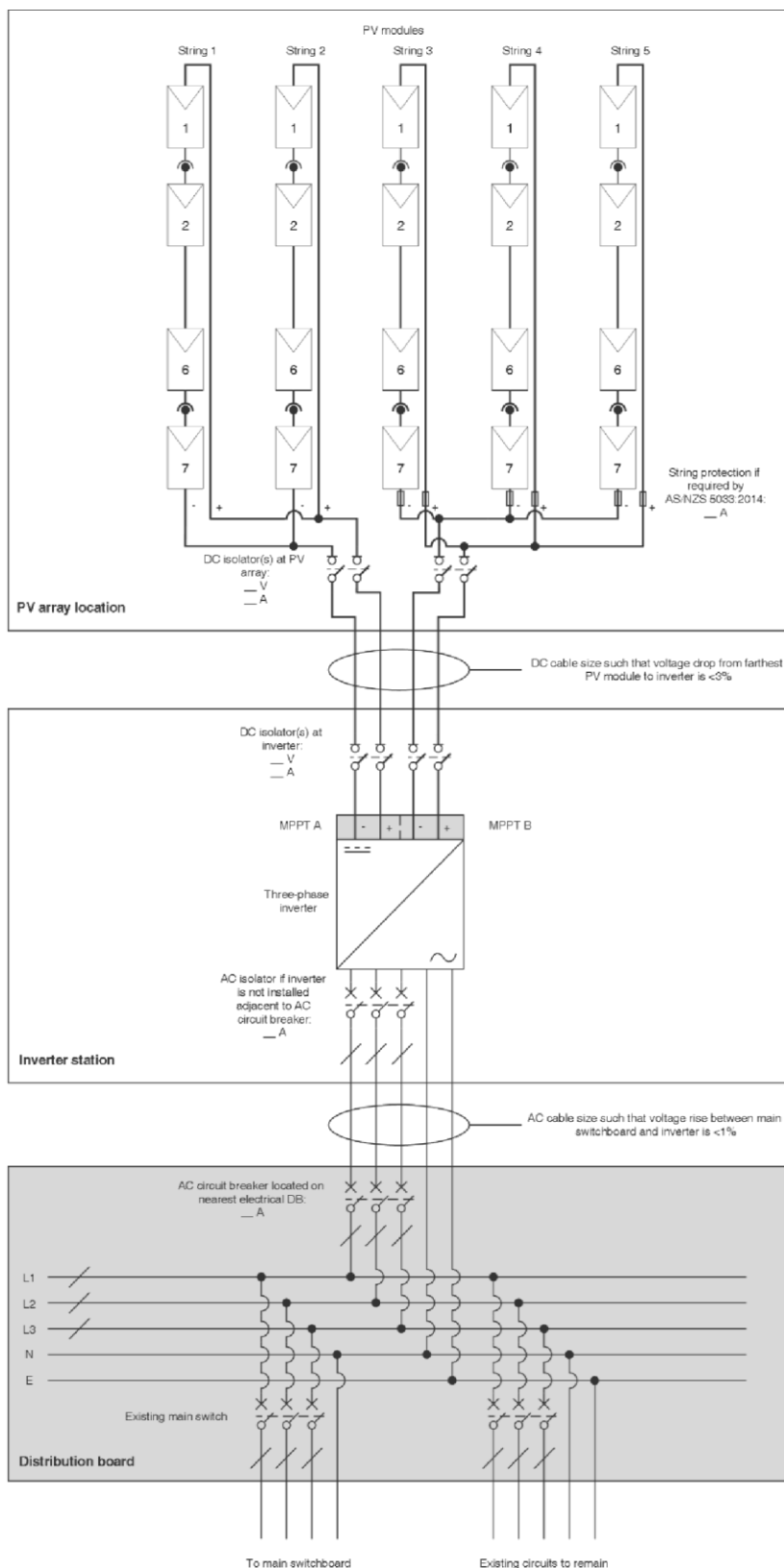
A permanent safe roof access system must be provided to facilitate ongoing inspection and maintenance of the system.

5.2.4 System Design

The design of the Solar PV system must meet requirements as set out by the available international standard, local requirement and guidelines.

To prevent circulating currents, reduce mismatch losses and improve Solar PV array energy yield, all Solar PV strings connected in parallel to the same inverter input must have less than 5% voltage mismatch and have similar rated electrical characteristics.

The designer of the Solar PV system must provide single line diagrams similar to **Figure 1**, including electrical ratings of all isolators and switches etc.

Figure 1: Example Single line diagram of three phase PV system

5.2.5 Location of Solar PV Array

The Solar PV modules must be arranged so that they are grouped together where possible and are symmetrically or geometrically ordered.

Modules connected in the same string must be arranged so that their tilt and orientation are within $\pm 5^\circ$ of each other. This does not include micro-inverters and DC power optimisers connected to individual modules.

The location of the Solar PV array should be designed to maximise annual solar generation:

- If flush mounting system is used, the array should be placed on roofs with a northerly aspect. All flush mounted arrays must achieve a minimum tilt of 10° from horizontal;
- If raked mounting system is used, the Solar PV modules should be inclined to the same angle as the angle of latitude of the site $\pm 10^\circ$ and oriented so that panels face due north;
- If a north-facing roof is not available, installing on east or west facing roofs is permissible provided the system generation estimate meets the specified system size.

5.2.6 System Size

The Solar PV system size must be provided as both the sum of the Solar PV array nominal capacity rating under Standard Test Conditions in kilowatt peak (kWp), and the sum of the nominal power output of all inverters in the system in kilovolt-ampere (kVA).

Unless otherwise specified by Micron, the Solar PV system capacity must be determined from the energy usage of the Micron building; the system provider must propose a system size and provide payback calculations. This information should reference the site's existing power bills, and interval meter data as the basis for these calculations, or for new buildings, an estimate of the predicted load profile at the connection point.

The expected operating life of the system should be at least 20 years. For the purposes of estimating the cost of ongoing system maintenance, assume that Micron will carry out a system operation and maintenance inspection every year. Ongoing system costs should include inverter replacement every 10 years.

To aid in the accuracy of the above, the system provider should obtain the following information from Micron:

- Electricity usage and load profile;
- Cost of electricity during the hours of generation

5.2.7 Power System Study

Power system study have to be conducted to determine the technical feasibility of the Solar PV connection based on latest network readings as well as finalized inverter and equipment parameters on the plant side.

The scope of study and assessment include the followings:

- a) Assessment of the technical impact of connection of the solar PV plant in terms of:
 - i Power flow Analysis
 - ii Short Circuit Analysis
 - iii Power Factor Analysis
 - iv System Adequacy – Penetration Limit
 - v Protection coordination study

5.2.8 Selection Of connection point

Wherever feasible, it is highly recommended the Solar PV power generation is connected only to non manufacturing load.

Note: *This to minimize risk of power quality related incident from the Solar PV generation affecting electrical system at FAB and CUB building*

The following guides shall be satisfied to ensure that the connection does not interfere with the existing power supplied.

The following items are to be considered during design.

- a) Load during peak and trough
- b) Anti-islanding
- c) Protection system
- d) Step-up transformer (if applicable)
- e) Interlocking
- f) Back-up power supply (if applicable)
- g) Energy storage system (if applicable)
- h) Sensitive load

Power quality and energy monitoring is required at the connection point/feeder and link to FMCS to monitor the performance of the PV solar system

Surge protection is required at the connection point/feeder.

5.3 System Equipment Requirements

5.3.1 Solar PV Module Requirements

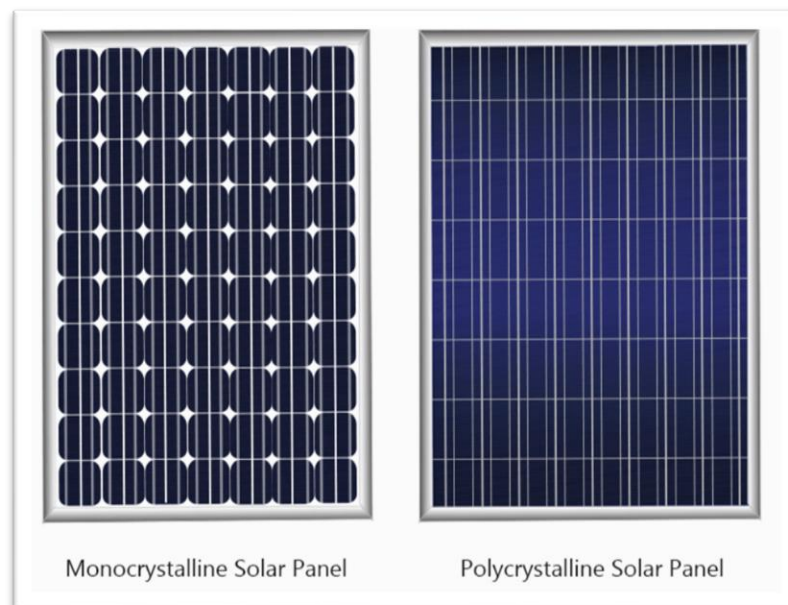
All PV modules proposed in a single PV system should have the same manufacturer and model.

5.3.1.1 Certifications

All Solar PV modules must be certified to IEC-61730. The Fire Test MST-23 must be carried out as part of the IEC 61730-2 testing and IEC 61730 certification. Modules installed on buildings must be listed as fire tested on the approved PV modules list at time of installation.

Crystalline photovoltaic modules (monocrystalline and multi/poly-crystalline): must be certified to IEC-61215

Figure 2: Example of solar module



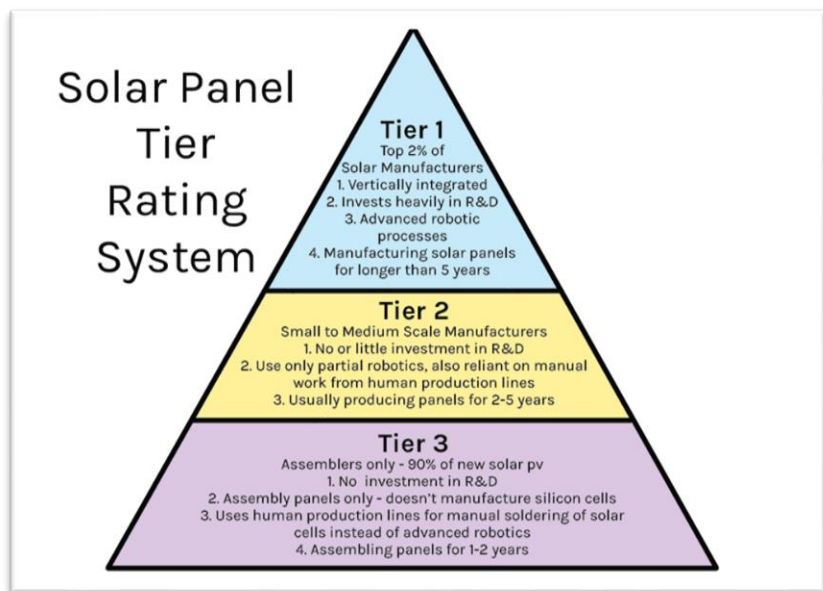
5.3.1.2 Solar PV modules recommended to be Tier 1 (minimum Tier 2) Technology / Manufacturer / Supplier Brand

Tier 1 : Considered to be the best solar panels. Manufacturers are vertically integrated; they have control over the entire manufacturing process. They use high-grade silicon solar cells and are the best performing solar panels.

Tier 2 : These panels don't necessarily have substandard quality compared to Tier 1, but tier 2 manufacturers don't control the whole process. They spend less on research and design, but they often still feature the latest developments on design and construction.

Tier 3 : These manufacturers are the largest percentage in the market. Often tier 3 businesses only assemble parts, and source their products on the open market.

Figure 3 : Solar Panel Tier Rating System



5.3.1.3 Protection Rating

Solar PV module junction boxes that are exposed to the environment must be at least IP65 compliant and must be UV resistant.

Solar PV modules installed in a coastal environment or areas with high agricultural activities must be certified to the following IEC standards respectively:

IEC 61701 Ed 2.0 Salt mist corrosion testing of photovoltaic (PV) modules

IEC 62716 Ed 1.0 Ammonia corrosion testing of photovoltaic (PV) modules

Preference should be given to PV modules certified to the above IEC standard in all installations.

5.3.1.4 Temperature Coefficient

Maximum allowable temperature coefficient of maximum power is -0.5% /°C.

5.3.1.5 Module Efficiency

Monocrystalline: > 18%

5.3.1.6 Operating Temperature

Operating module temperature -20°C to +80°C.

5.3.1.7 Connectors and Cables

Connectors must be compliant with EN 50521 (2012). Datasheet and/or installation manual must indicate the make and model of the connector.

5.3.1.8 Warranty

Minimum manufacturer's workmanship Warranty: 10 years

Minimum manufacturer's power warranty: 10 years at 90% power output and 25 years at 80% power output or better.

5.3.2 Inverter Requirements

5.3.2.1 Selection of Inverter

Transformerless inverters are not to be used if the Solar PV module selected requires functional earthing (i.e. connection of one of the Solar PV array conductors to earth).

Inverters installed should have Ingress Protection of IP 54 for indoor mounting, IP65 for outdoor mounting (fitted with a sun cover/protected canopy when appropriate)

5.3.2.2 Certification

The inverter supplied must be compliant to IEC 62109-1 and IEC 62109-2.

5.3.2.3 Technical Requirements

The inverter must be provided with a Modbus output, and protocol converter (if required) to facilitate a wired connection of the inverter to the FMCS.

The inverter must meet all requirements as specified by known international standard including the following requirements:

Total output power (AC)	To match solar PV plant capacity while achieving optimum system efficiency
Input DC voltage range	As required for the solar grid inverter DC input
Maximum power point (MPPT)	Shall be incorporated
Number of independent MPPT inputs	1 or more
Operation AC voltage	For up to 5kWp – Single phase For above 5kWp – Three phase
Operating Frequency range	47.5 - 52.5 Hz 57.5 – 62.5H
Nominal	50 Hz

Frequency	60 Hz
Power factor of the inverter	>0.98 at nominal power
Total harmonic distortion	Less than 3 %
Built-in protection	AC high/low voltage, AC high/low frequency
Anti-islanding protection	As per IEC 60255.5/ IEC 60255.27 / IEC 62116
Operating ambient temperature range	-1°C to +55°C
Humidity	0 - 95% Rh
Inverter efficiency	>= 95%
Inverter weighted efficiency	>= 94%
Protection degree	IP65 for outdoor mounting, IP 54 for indoor mounting
Communication interface	RS485/RS232/RJ45/
Safety compliance	IEC 62109-1, IEC 62109-2
Environmental Testing	IEC 60068-2 (1,2,14,30)
Efficiency Measurement Procedure	IEC 61683
Cooling	Convection
Display type	LCD for data display, LCD/LED for status display
Display parameters to include (minimum)	Output power (W), cumulative energy (Wh), DC voltage (V), DC current (A), AC voltage (V), AC frequency (Hz), AC current (A), cumulative hours of operation (h)

- The inverter must be capable of autonomously synchronizing with the existing grid voltage.
- Anti-Islanding protection: both active and passive protection required.
- Earth fault alarm system must be provided to IEC 62109-2 ed. 1 13.9 fault indication requirements
- The earth fault alarm should be installed such that any fault indication is detectable by the system owner and/or FMCS.

5.3.2.4 Other requirement

DC connectors between the inverter and Solar PV array DC input must be of the same manufacturer and model.

5.3.2.5 Warranty

The inverter must have a minimum warranty of 10 years.

5.3.3 Mounting System Requirement

The mounting frame and associated parts, including but not limited to parts such as bolts, splices, etc. must be a suitable proprietary made product. Custom made products are not acceptable.

The mounting system manufacturer's installation specification, including but not limited to: maximum fixing width for wind regions and roof zones, maximum building height and roof mounting requirements, must be followed.

5.3.4 Monitoring System Requirement

The system must be integrated to the Micron's FMCS in accordance with the Micron's FMCS System Standard. The system monitoring cover as a minimum;

1. Electrical meters at the distribution board
2. Inverter monitoring including;
 - a. DC String voltages
 - b. DC String currents
 - c. DC Power
 - d. AC voltage
 - e. AC current
 - f. AC power
 - g. AC kVA
 - h. AC power factor
 - i. THD
 - j. Earth fault detection and alarming

If weather monitoring is required, sensors and data logging for insolation (measured at array tilt) and Solar PV cell temperature must be provided.

If energy storage is installed, monitoring and data logging of charging, discharging and state of charge must be provided.

All system monitoring data must be collectively recorded so that the entire system, including Solar PV production, weather data and energy storage data, can be easily assessed.

Minimum to generate daily, monthly and yearly solar power generation reports, charts and trends (remote, web based applications, etc)

5.3.5 Balance of System Component Requirements

All balance of system components installed on the DC side of the system must be rated for DC application.

All equipment, including any conduit and ducting, exposed to the outdoor environment must be a minimum of IP55 compliant and must be UV resistant.

5.3.5.1 DC Cables

Solar PV cables must comply with requirements set out by IEC 62930.

Cable size must be selected such that the voltage drop from furthestmost the Solar PV module to the inverter is less than 3%.

All AC and DC wiring and components must be installed in separate enclosures and run in separate conduit.

5.3.5.2 DC isolators

DC isolators must be installed where required by IEC 60364-7-712, and the isolators must meet the device requirements set out by IEC 60364-7-712.

DC isolators should meet the following specifications:

- Type: Rotary type DC isolator;
- IP rating: IP66 or higher;
- Provide within specification sheet:
 - a. Single pole voltage and current rating
 - b. Wiring diagram

Isolator enclosures must be installed according to manufacturer's specifications to ensure IP rating is maintained.

DC isolator enclosures should be constructed of metal and equipotentially bonded

5.3.5.3 AC cable

AC cable between Solar PV system inverter output and consumer mains must be selected such that:

- Voltage rise between inverter and main switchboard is less than 1% and,
- Voltage rise between main switchboard and consumer main is less than 1%.

5.3.5.4 AC circuit breaker

AC circuit breaker acting as the main switch (inverter supply) must be rated to protect the cable against overcurrent to IEC 60947-2.

AC circuit breaker ratings must be above inverter maximum AC current output rating to avoid unnecessary tripping of circuit breaker. The cable between the inverter and the AC circuit breaker must have a higher current carrying capacity than the AC circuit breaker rating and the maximum AC current output of the inverter.

5.3.5.5. Solar PV string protection

Solar PV string overcurrent protection, if needed, must adhere to sizing and location requirements as set out by IEC 60364-7-712 2017 712.433.1.101.2

5.3.5.6 Plugs, sockets and connectors

Plugs, sockets and connectors used in the Solar PV system must comply with IEC 62852 "Connectors for DC-application in photovoltaic systems - Safety requirements and tests"

Note: Plugs and sockets must only be mated with those of the same type from the same manufacturer. Mating of plugs and sockets that are compatible but do not meet the above criteria is not acceptable.

Ensure that Solar PV modules and/or inverter warranty will not be voided in the event that pre-installed connectors need to be replaced. To avoid the need to replace pre-installed connectors it is recommended to only use Solar PV modules and inverters for which matching connectors can be sourced.

5.3.5.7 Low Voltage Switchboard

Refer to the following document: [Global Facilities - Design & Construction Standard - Electrical - Electrical Specification](#)

Section 6.11: Low Voltage Switchgear and Control Gear
The degree of protection as per IEC 60529 for any assembly shall be at least:

- For indoor use (in enclosed buildings) IP 31
- For outdoor use (protected by e.g. rain canopy) IP 55

5.3.5.8 Cable Ladder/Tray

Refer to the following document: [Global Facilities - Design & Construction Standard - Electrical - Electrical Specification](#)

- Section 6.17.2 Cable Ladder/Tray
Cable trays on exposed roof shall be covered to protect cables from ultra-violet radiations.

5.4 System Installation

Solar PV system installation includes the installation of all PV system components, e.g. PV modules, inverters and balance of system components, as well as ancillary work such as forming openings in the building envelop for cable entries etc. The Solar PV system provider must ensure that the integrity of the building, especially waterproofing and fireproofing properties, is not affected by the installation of the Solar PV system.

All Solar PV system components must be installed in a manner such that the manufacturers' product warranties are not affected.

5.4.1 Asset Labelling

Equipment must be provided with asset labels.

5.4.2 Solar PV Module Installation

Installation of the Solar PV modules onto the mounting system and attachment of mounting system equipment to the roof must follow the respective manufacturer's specifications for the specific application.

All equipment used must be fit for purpose and care is to be taken while installing so as to not adversely affect the system finish or void warranties.

Ensure that bolts, washers, screws, nails or other metal objects in contact with the mounting frame are of a similar metal to reduce the likelihood and severity of galvanic reaction.

5.4.2.6 Module Performance

Solar PV modules must be installed such that all strings connected in parallel to the same MPPT input comprise the same number of panels and are of the same make and model.

A string of Solar PV modules must be connected such that all modules in the string have the same tilt and the same orientation.

5.4.2.7 Mounting Methods

Solar PV modules can be installed or fixed by Bolts method or by clamp method. For bolts type fixing corrosion resistive M8 bolts to be used/ Mounting methods should ensure following things:

- Minimum clearance required between module frame and surface of the roof or wall is 200mm.
- Minimum distance between each Solar PV Module should be 10.50mm.
- Drainage holes should be kept open and no blockage is allowed in the drainage hole.

Solar PV modules are not to be subjected to wind loads exceeding the maximum permissible loads and should not be subjected to excessive forces due to the thermal expansion of support structures , careful consideration has to be shown during system design and installation such that thermal expansion of support structures do not cause any breakage of Solar PV modules which will not be part of Warranty Document.

5.4.2.8 Example installation mounting Non-FAB Rooftop

a) Reinforced Concrete Roof

Principle of Installation:

- Method A by cast Concrete Stump. See **Figure 4**
- Method B by Ballast-Weighted mounting structure. See **Figure 5**
- Both methods act as counterweight to hold the panel to the ground.
- No penetration to roof to ensure water tightness and integrity of the roof.
- Waterproofing felt is placed underneath the concrete stump as additional layer of protection for existing waterproofing membrane.

- The installation requires structural PE endorsement.

b) Metal Roof

Principle of Installation:

- The Klip-lok clamp will be used for installation. See **Figure 6a** and **Figure 6b**
- Clamp is fixed on the steel tile with bolt.2
- No penetration to roof to ensure water tightness and integrity of the roof.

Figure 4: Example Concrete Stump structure



Figure 5: Example Ballast-Weighted Mounting Structure



Figure 6a : Example Installation Using Klip-lok



Figure 6b: Example Installation Using Klip-lok



5.4.2.9 Example installation mounting on FAB Rooftop

a. Bondek Slab Roof

Principle of Installation:

- Installation by cast Concrete Stump.

See **Figure 7**

- Concrete stump acts as counterweight to hold the panel to the ground.
- No penetration to roof to ensure water tightness and integrity of the roof.
- Waterproofing felt is placed underneath the concrete stump as additional layer of protection for existing waterproofing membrane.
- The installation requires structural PE endorsement

b. Membrane Roof

Principle of Installation:

- Method A: by Aerocompact structures – low profile mounting solutions which has similar stability and reliability as installation method used for RC roof. See **Figure 8**
- Method B by UniAnchor – the PVC part of the UniAnchor is welded to the roofing membrane. See **Figure 9**
- No penetration to roof to ensure water tightness and integrity of the roof.

5.4.2.5 Solar PV module spacing

A gap should be provided between Solar PV modules to allow for thermal expansion.

If excess roof space is available, large arrays should be installed as follows so that adequate access is provided for maintenance of the array:

- 500mm clearance around the perimeter of the Solar PV array;
- For flush mounted arrays, every fourth row is spaced at least 700mm to provide a maintenance access way;
- Ensure that access to individual PV modules does not require the removal of more than one other Solar PV module.

Figure 7: Example Concrete Stump Installation



Figure 8: Example Aerocompact structures



Figure 9: Example UniAnchor



5.4.2.5 Solar PV module spacing

A gap should be provided between Solar PV modules to allow for thermal expansion.

If excess roof space is available, large arrays should be installed as follows so that adequate access is provided for maintenance of the array:

- 500mm clearance around the perimeter of the Solar PV array;
- For flush mounted arrays, every fourth row is spaced at least 700mm to provide a maintenance access way;
- Ensure that access to individual PV modules does not require the removal of more than one other Solar PV module.

5.4.3 Inverter Installation

The area surrounding the inverter and all associated protective devices such as Solar PV array DC isolator, inverter AC isolators, and wiring must be known as the inverter station.

In addition, where multiple inverters are installed connecting to the same switchboard, the inverters must be installed grouped in a common location. Each inverter must be provided with an AC isolation switch at an electrical distribution board. A main switch (inverter supply) must be provided to isolate all inverters installed on the distribution board.

5.4.3.1 Inverter Station Location Selection

Inverter station location must have restricted access. General public must not be able to gain unauthorised access.

Inverters must be installed such that the ventilation and clearance requirements recommended by the inverter manufacturer are met.

The inverter station should be located and configured such that:

- No inverter is exposed to excessive heat, dust or moisture;
- No inverter is exposed to direct sunlight or rain;
- DC losses are minimised by locating the inverter as close as possible to the Solar PV array;

- AC losses are minimised by locating the inverter as close as possible to the nearest electrical distribution board;

5.4.4 Cable Management/Wiring

Regardless of system voltage, the entire Solar PV installation, including the Solar PV array, inverters, associated wiring and protection, must have restricted access.

Conduits must be installed up to and including the cables at the inverter input. The conduits must be supported in such a way that it cannot be subjected to mechanical strain.

All DC cabling from the inverters to the Solar PV array DC isolators adjacent to the array must be installed in metal or heavy duty (HD) conduit. This conduit must also be UV resistant (marked 'T'). The conduit must be labelled "Solar" every 2 metres, with the labelling clearly visible when installed. 'HD' and 'T' markings on the conduit should also be visible.

Cable installation must not be susceptible to damage by vermin or birds.

Cables must be installed:

- Cables must not lay on roofs or floors without adequate enclosure;
- Plastic cable ties must not be used.
- DC connections must not be hanging under tension

5.4.5 Solar PV System Signage

All Solar PV system components must be labelled. Signs installed should be engraved plastic or metallic labels.

In addition:

- Provide a plan view of Solar PV system. The plan view should show the location of the Solar PV modules, DC isolators, inverter, point of connection.
- Label terminology must be identical to the terminology used in the system's shutdown procedure and other required documentation;
- A list of action/s to be taken if the earth fault alarm is activated should be placed next to the earth fault alarm and/or given to those who will receive earth fault notifications (Micron facilities); where the Solar PV system contains multiple Solar PV arrays and/or inverters,

provide labelling on all isolators, inverters and arrays, identifying corresponding system components. For example, "Array - A" is connected to the rooftop "PV Array DC isolator - A", inverter "PV Array DC isolator - A", "Inverter - A" and so on.

5.4.6 Lightning Protection

Lightning protection with dedicated earthing shall be provided to protect the PV installation.

The metal portions of photovoltaic array should be bonded to the lightning protection system.

5.4.6.1 Surge Protection Devices (SPD)

SPDs installed on the DC side of the Solar PV system must be explicitly designed and manufactured for DC PV application.

Surge protection should be installed for each inverter. Surge protection to meet the following guideline:

- Connection type: shunt diverter
- Nominal voltage: > Solar PV system operating DC voltage
- Maximum continuous voltage: 1000VDC
- Maximum continuous current rating: Solar PV system short circuit current
- Response time: < 5nS
- Maximum surge current (8/20μS): > 20kA
- Earth leakage current: <10mA

5.4.6.2 Minimizing wiring loops

To reduce the magnitude of lightning-induced over voltages, the Solar PV array wiring should be laid in such a way that the area of conductive loops is minimized.

5.4.7 Earthing

Separate dedicated earthing shall be provided to protect the PV installation.

All exposed conductive parts of the Solar PV system, including Solar PV module frame and mounting rails, must be bonded to earth (equipotential bonding).

Both mounting rails on every row of Solar PV modules must be bonded to earth.

The removal of any one module must not affect the bonding of the system. To ensure that bonding is maintained across rail joins, earthing straps must be installed across rail splices unless the manufacturer confirms (in documentation) that the splices maintain continuity.

5.4.8 Water Supply

Provision of water supply at roof level is to be provided for maintenance/cleaning of solar PV module

6.0 Commissioning

See Appendix A: Solar PV System Commissioning for more information.

7.0 Operations and Maintenance

An example Maintenance procedure and timetable has been provided in Appendix B: Solar PV System Maintenance. This is to be used if the procedure provided by the system provider is not available

8.0 Document Control

8.1 **Approval:** GLOBAL_FAC_GFTT_APPR

8.2 Notification:

GLOBAL_FAC_MANAGERS
 GLOBAL_FAC_COMMUNICATIONS
 GLOBAL_FAC_CONSTRUCTION
 GLOBAL_FAC_NOTIFY
 FCG_F2_FAC_NOTIFY
 FCG_F4_FAC_NOTIFY
 FCG_F6_FAC_NOTIFY
 FCG_F10W_FAC_NOTIFY
 FCG_F10N_FAC_NOTIFY
 FCG_MTTW_FAC_NOTIFY
 FCG_F15_FAC_NOTIFY
 FCG_F16_FAC_NOTIFY
 FCG_MMP_FAC_NOTIFY
 FCG_MMY_FAC_NOTIFY
 FCG_MSB_FAC_NOTIFY
 FCG_MXA_FAC_NOTIFY
 MTB_FAC_ALL_NOTIFY

8.3 Final Viewer Access

All Micron Team Members at all Sites

8.4 Retention

Adherence to the requirements specified in the
[**Global Facilities - Document Control Procedures**](#)

8.5 Review

Biennially (two years)

9.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	07/02/21

10.0 APPENDIX

APPENDIX A: Solar PV System Commissioning

The Solar PV system installer must commission Solar PV system once installation has been completed. The documented results of the system commissioning will confirm that the system meets the system design and is structurally and electrically compliant with the relevant standards and guidelines.

The commissioning requirements as per standards IEC 62446-must be met as a minimum.

As part of system commissioning, the installer will verify that the system is installed correctly using an installation checklist to document this process. The installer will then test the system and record the results in a test record document. The installation checklist and test records together form the system commissioning sheets. Micron requires the Solar PV system's completed commissioning sheets are to be included in the system documentation given to the Solar PV system owner.

The following pages contain an example of an installation checklist and test record suitable for use as part of the system commissioning.

Example Installation Checklist

Installer name:	
Installer accreditation number:	
Installer signature:	
Commissioning date:	
Installation address:	
PV module manufacture:	PV module model number:
PV modules comply with relevant building codes: YES ☐ NO ☐	
All PV modules connected to the same MPPT are of the same make and model or have similar rated electrical characteristics: YES ☐ NO ☐	
All PV modules connected to the same string have the same angle of altitude and azimuth: YES ☐ NO ☐	

Number of modules in series: String 1: String 2: String 3: String N:	Number of strings in parallel: Sub-Array 1: Sub-Array 2: Sub-Array 3: Sub-Array N:
Array mounting system manufacturer:	
Array mounting system type:	
Array mounting system certified for installation site parameters: YES ☐ NO ☐	
Array mounting system does not use or contact any galvanically dissimilar metals: YES ☐ NO ☐	
All penetrations and fixings are suitably sealed and weatherproofed: YES ☐ NO ☐	
PV array voltage complies with site regulations: YES ☐ NO ☐	
PV system wiring is suitably protected from mechanical action: YES ☐ NO ☐	
PV array uses single core double insulated cabling compliant to relevant standards: YES ☐ NO ☐	
Overcurrent protection is provided where required: YES ☐ NO ☐ NOT REQUIRED ☐	
All components used are rated for DC usage and have voltage ratings greater than or equal to the PV array maximum voltage: YES ☐ NO ☐	
All components are rated suitable to their environment and have the appropriate IP and UV ratings: YES ☐ NO ☐	
Disconnection devices and protection devices (where installed) are readily available in the case of maintenance or emergency: YES ☐ NO ☐	
Disconnecting devices comply with frequency of usage requirements and are rated for the temperature adjusted operational circuit current: YES ☐ NO ☐	

Switch-disconnectors are rated to switch full load currents, are not polarity sensitive and grade with overcurrent protection (where required):
YES ☐ NO ☐

PV array switch-disconnectors interrupt all live conductors:
YES ☐ NO ☐

PV conductor current carrying capacity is equal to or greater than: (i) the potential system fault current or (ii) the overcurrent protection (where installed):
YES ☐ NO ☐

PV cabling, where exposed to the elements, is UV-resistant or installed in UV-resistant enclosures:
YES ☐ NO ☐

A method of securing cabling has been used which will last the lifetime of the system:
YES ☐ NO ☐

DC cabling within buildings is enclosed in heavy-duty rated protection: YES ☐ NO ☐ NOT REQUIRED ☐

Combiner boxes are installed according to manufacturer recommendations: YES ☐ NO ☐

Combiner boxes are suitably protected from the environment using appropriate bottom entry cable glands:
YES ☐ NO ☐

Double insulation between all conductors is maintained throughout the system: YES ☐ NO ☐

PV plugs, sockets and connectors comply with relevant standards, are rated for the installation environment and only connect with the same make and model:
YES ☐ NO ☐

Blocking and bypass diodes, where installed external to the PV module are suitably protected and grade according to relevant standards and system parameters: YES ☐ NO ☐

Overcurrent protection (where required) is installed at the end of the conductor which is most electrically remote from the PV modules:
YES ☐ NO ☐ NOT REQUIRED ☐

Switch-disconnector is mechanically interlocked with or located adjacent to the inverter if the array is not in line of site or if it is greater than 3 metres away:
YES ☐ NO ☐

Where multiple disconnection devices are installed they are ganged together or grouped and labelled such that it is clear all must be operated to isolate the system: YES ☐ NO ☐

All exposed metal module frames and mounting equipment are earthed and equipotentially bonded in accordance with the relevant standards: YES ☐ NO ☐

Equipment used for PV module and mounting frame earth connections is fit for purpose: YES ☐ NO ☐

Earthing has been arranged so that the removal of a single module earth connection will not disrupt the continuity of the bonding connections for the rest of the array: YES ☐ NO ☐

Earthing conductor type and size comply with the relevant standard: YES ☐ NO ☐

PV array functional earthing is done close to or within the inverter and is done according to the relevant standards: YES ☐ NO ☐ NOT REQUIRED ☐

The inverter complies with the relevant country standards: YES ☐ NO ☐

The inverter installation complies with manufacturer's instructions, DNSP rules and regulations and relevant state legislation: YES ☐ NO ☐

Inverters connected to LV PV arrays have an internal or external earth fault alarm system: YES ☐ NO ☐

Inverters connected to arrays with direct functional earthing have an earth fault interrupter which will shut the PV system down and provide a fault alarm when an earth fault occurs: YES ☐ NO ☐ NOT REQUIRED ☐

Signage and labelling conforms to the relevant standards and guidelines: YES ☐ NO ☐

Example Test Records

PV Array DC:

If array is LV (>120 V) there is no voltage on input side of array combiner box (if one is installed)	☐
There is no voltage on output side of array combiner box (if one is installed)	☐
Continuity between strings and array combiner box:	
String 1 +ve	☐
String 1 -ve	☐
String 2 +ve	☐
String 2 -ve	☐
String 3 +ve	☐
String 3 -ve	☐
String 4 +ve	☐
String 4 -ve	☐
Correct polarity between strings and array junction box:	
String 1	☐
String 2	☐
String 3	☐
String 4	☐
Open circuit voltages:	
String 1	V
String 2	V
String 3	V
String 4	V
Continuity between array combiner box and PV array DC isolator:	
Sub-Array +ve	☐
Sub-Array -ve	☐
Correct Polarity between Array Junction Box and PV array DC isolator	☐
Sub-Array Voc	V
Short Circuit Currents:	
String 1	A
String 2	A
String 3	A
String 4	A
Short Circuit Current Array	A

<i>Reconnect the strings one at a time by reconnecting the string overcurrent protection (either fuses or circuit breakers) and connecting any disconnectors.</i>	
Open Circuit Voltage at input side of PV array DC isolator	V
Continuity between PV array DC isolator and Inverter:	
Array +ve	☐
Array -ve	☐
Correct Polarity between PV array DC isolator and Inverter	☐

<u>Inverter AC:</u>	
Continuity between Inverter and kWh meter:	
Line	☐
Neutral	☐
Correct polarity between inverter and kWh meter	☐
Continuity between kWh meter and PV inverter AC isolator:	
Line	☐
Neutral	☐
Correct Polarity between kWh meter and PV inverter AC isolator	☐
Correct Polarity at output of PV inverter AC isolator from grid	☐
Voltage at output of PV inverter AC isolator from grid	V
Initial reading of kWh meter	

<u>Start-up of system:</u>	
<i>Refer to system manual for the inverter and follow start-up procedure. This generally involves turning on the PV array DC isolator followed by inverter AC isolator, but the procedures as recommended by the inverter manufacturer must be followed.</i>	
System connects to grid	☐
<i>When inverter AC isolator and PV array DC isolator turned on and inverter start-up procedure followed</i>	
Voltage at DC input of inverter	V
Voltage within operating limits of inverter	☐
Voltage at AC output of inverter	V
Input power of the inverter (If available)	W
Output power of the inverter (If available)	W
Output power as expected	☐
System disconnects from grid when inverter AC isolator turned off	☐

APPENDIX B : Solar PV System Maintenance

All maintenance on the PV system should be conducted in accordance with documentation provided by the system provider. In the event that one does not exist the following has been produced and must be implemented as a minimum.

Activity	Frequency
Visually inspect PV modules - Defects, such as yellowing, micro cracks and hotspots; - Dust/dirt build-up.	Annually
Check mechanical integrity of the array structure - Screw and boards are secure; - No signs of cracks, corrosion or other weakness.	Annually
Clean modules - Using water only; - Avoid directly spraying water onto cabling, junction boxes and PV array isolator enclosures.	As required (dependent on site)
Inspect and clean inverter - Clearance for ventilation is maintained - Any ventilation vents are clear - Inverter is securely mounted to the wall	Annually
Check all cabling and conduit: - Mechanical damage - UV damage	Annually
Check output voltage and current of each string of the array and compare to the expected output under the existing conditions.	Annually
Check electrical wiring for loose connections	Annually
Check the operation of the isolators and circuit breakers. - Rooftop and inverter PV Array DC Isolators; - Inverter AC Isolation; - Solar supply main switch.	Annually
Check monitoring system: - Logging production data; - Functioning network connection for remote monitoring and earth fault notification (if applicable).	Annually

APPENDIX C: Documentation Checklist

Below are the list of minimum documentation required for every projects.

Drawings:

1. A plan view of the Solar PV array, clearly showing:
 - a. How the Solar PV array is wired;
 - b. Access routes (as relevant);
 - c. Electrical reticulation;
 - d. Penetration points;
 - e. Module identifier number
 - f. Tilt angle
 - g. Location of key system equipment and switches.
2. Single line diagram/s, in accordance with the showing;
 - a. connection of all Solar PV system equipment to electrical distribution board, including the electrical ratings of the Solar PV array and electrical ratings of all overcurrent devices, isolators and switches
 - b. A table nominating which strings are connected to which inverter, by both inverter number and inverter input (A/B);
 - c. Indicate total number of panels and installed capacity on title block;
 - d. Lightning protection system (if required);
 - e. String configuration back to the inverters. String configuration is to be identified using hatching or colour and string number to distinguish strings;
 - f. Location and rating of DC isolation points and the strings connected;
 - g. DC string fuses including rating (if required);
 - h. Surge arrestor;
 - i. Cable size;
 - j. Identifying AC connection point (DB name/number) and breaker size;
 - k. Inverter schedule;
 - l. Indicating type and number of panels connected to each inverter;
 - m. Array Voc;
 - n. Array Isc;
 - o. Inverter maximum AC current rating;
 - p. Panel Voc;
 - q. Panel Isc;
 - r. Secondary protection devices;
3. Earthing diagram
4. Metering and monitoring drawings:
 - a. Power;
 - b. Communications;

5. Structural drawings:

- a. Mounting system drawings and details as required;
- b. Details of any structures that tie into the building fabric. e.g. structural plinths.

Operation and Maintenance Manual, including:

1. A description of the function and operation of the system's installed equipment;
2. A procedure to verify the correct operation of the system;
3. The supplier's contact details for installation queries and system support;
4. The shutdown and isolation procedure for emergency and maintenance, including any electrical safety warnings;
5. A list of equipment supplied, with serial numbers of all equipment where applicable;
6. A list of actions to be taken in the event of an earth fault alarm;
7. A site specific system performance estimate providing monthly generation estimates taking into account expected seasonal, operational and site specific variation. The system performance estimate should at a minimum factor in effects of the Solar PV array orientation, shading effects and de-rating factors such as temperature and equipment efficiency. Any assumptions made must be clearly stated.
8. Array frame engineering certificate for wind and mechanical loading;
9. Installer/designer's declaration of compliance to available local standard

Mechanical design;

10. Warranty information for all applicable equipment and workmanship warranties;
11. Network Provider's Solar PV system approval to connect and connection offer;
12. Equipment manufacturer's documentation and handbooks for all equipment supplied. As a minimum the following must be included:
 - a. Panels;
 - b. Mounting frame;
 - c. Inverter;
 - d. Isolators;
 - e. Cable;
 - f. Monitoring devices.
13. Log-in details for any monitoring systems
14. The commissioning sheet and installation checklist;